(ii) Let $\cos^{-1}\left(\frac{4}{5}\right)=\theta$. Then $\cos\theta=\frac{4}{5}$

$\Rightarrow \sin\theta=\frac{3}{5}$

$\therefore \tan\theta=\frac{3}{4} \Rightarrow \theta=\tan^{-1}\left(\frac{3}{4}\right)$

$\therefore \cos^{-1}\left(\frac{4}{5}\right)=\tan^{-1}\left(\frac{3}{4}\right)$

Now, $\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{3}{5}\right)$

$=\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{3}{5}\right)$

$=\tan^{-1}\left(\frac{\frac{3}{4}+\frac{3}{5}}{1-\frac{3}{4}\times\frac{3}{5}}\right)$

$=\tan^{-1}\left(\frac{27}{11}\right)$

Example 4.26

Find the value of $\tan\left[\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{8}\right)\right]$

Solution

Let $\tan^{-1}\left(\frac{1}{8}\right)=\theta$

Then $\tan\theta=\frac{1}{8}$

$\therefore \tan\left[\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{8}\right)\right]=\tan\left(\frac{\pi}{4}-\theta\right)$

$=\frac{\tan\frac{\pi}{4}-\tan\theta}{1+\tan\frac{\pi}{4}\tan\theta}$

$=\frac{1-\frac{1}{8}}{1+\frac{1}{8}}=\frac{7}{9}$

Example 4.27

Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

Solution

$\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)$

$=\tan^{-1}\left[\frac{\frac{x-1}{x-2}+\frac{x+1}{x+2}}{1-\left(\frac{x-1}{x-2}\right)\left(\frac{x+1}{x+2}\right)}\right]=\tan^{-1}\left(\frac{2x^2-4}{-3}\right)$

Given that

$\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

$\therefore \tan^{-1}\left(\frac{2x^2-4}{-3}\right)=\frac{\pi}{4}$

$\frac{2x^2-4}{-3}=\tan\frac{\pi}{4}=1$

$2x^2-4=-3$

$\Rightarrow 2x^2-1=0$

$\Rightarrow x=\pm\frac{1}{\sqrt{2}}$

Example 4.28

Simplify: $\sin^{-1}\left(\frac{1}{3}\right)+\sin^{-1}\left(\frac{2}{3}\right)$

Solution

We know that

$\sin^{-1}(x)+\sin^{-1}(y)=\sin^{-1}\left[x\sqrt{1-y^2}+y\sqrt{1-x^2}\right]$

$\therefore \sin^{-1}\left(\frac{1}{3}\right)+\sin^{-1}\left(\frac{2}{3}\right)=\sin^{-1}\left[\frac{1}{3}\sqrt{1-\frac{4}{9}}+\frac{2}{3}\sqrt{1-\frac{1}{9}}\right]$

$=\sin^{-1}\left[\frac{1}{3}\sqrt{\frac{5}{9}}+\frac{2}{3}\sqrt{\frac{8}{9}}\right]$

$=\sin^{-1}\left(\frac{\sqrt{5}+4\sqrt{2}}{9}\right)$

Example 4.29

Solve $\tan^{-1}(x+2)+\tan^{-1}(2-x)=\tan^{-1}\left(\frac{2}{3}\right)$

Solution

$\tan^{-1}\left[\frac{(x+2)+(2-x)}{1-(x+2)(2-x)}\right]=\tan^{-1}\left(\frac{2}{3}\right)$

$\Rightarrow \tan^{-1}\left(\frac{4}{1-(4-x^2)}\right)=\tan^{-1}\left(\frac{2}{3}\right)$

$\Rightarrow 2x^2-6=12$

$\Rightarrow x^2=9$

$\therefore x=\pm 3$

Example 4.30

If $\tan(x+y)=42$ and $x=\tan^{-1}(2)$, then find y

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